Table 4
IGF-1 continuous reference values (ng/mL) using LMS[a] data obtained using the cobas e801 (Roche) immunoassay system at Yongin Severance Hospital.

Boys	No.	Percentiles							Girls	No.	Percentiles						
		2.5	10.0	25.0	50.0	75.0	90.0	97.5			2.5	10.0	25.0	50.0	75.0	90.0	97.5
Age, years									Age, years								
0	0	–			–			–	0	0	–			–			–
1	2	22.208	34.781	46.773	60.737	75.262	88.751	104.239	1	2	14.495	36.972	62.048	91.342	120.751	147.009	176.049
2	0	27.451	40.822	53.854	69.350	85.793	101.326	119.436	2	2	29.987	51.229	71.207	94.012	117.280	138.538	162.595
3	4	33.594	47.764	61.842	78.903	97.356	115.081	136.074	3	2	47.691	64.381	80.238	98.755	118.120	136.211	157.111
4	12	41.204	56.158	71.226	89.776	110.176	130.080	154.017	4	6	57.487	72.519	87.099	104.494	123.101	140.849	161.769
5	11	50.633	66.327	82.279	102.142	124.286	146.186	172.892	5	5	62.685	78.288	93.874	113.024	134.137	154.844	179.924
6	21	61.270	77.863	94.848	116.212	140.323	164.476	194.332	6	13	67.321	84.755	102.860	125.996	152.579	179.682	213.802
7	33	71.772	89.789	108.442	132.219	159.477	187.222	222.096	7	21	76.776	96.043	116.685	144.003	176.669	211.345	256.898
8	34	80.589	100.920	122.412	150.430	183.358	217.703	261.980	8	32	90.744	111.254	133.600	163.859	201.139	242.064	297.970
9	39	87.662	111.125	136.698	171.147	213.123	258.497	319.213	9	34	103.525	126.561	152.441	188.851	235.944	290.551	370.145
10	29	95.692	122.501	152.650	194.716	248.097	308.256	392.488	10	25	120.539	147.977	180.065	227.614	293.517	376.498	510.887
11	21	108.824	138.615	172.748	221.530	285.372	359.857	468.491	11	14	152.061	183.873	221.567	278.719	361.021	470.507	663.453
12	14	128.971	161.313	198.482	252.034	323.131	407.731	534.488	12	6	200.455	232.779	269.724	323.602	397.858	492.335	650.917
13	7	155.478	190.151	229.771	286.736	362.555	453.472	591.665	13	5	248.630	275.744	304.918	344.563	394.611	452.195	537.249
14	6	187.455	224.443	266.343	326.213	405.652	501.023	646.960	14	1	278.117	296.750	315.629	339.574	367.369	396.499	434.922
15	2	224.340	263.816	308.166	371.124	454.299	554.072	707.329	15	0	288.837	299.749	310.284	322.954	336.773	350.332	366.928
16	2	267.032	309.139	356.055	422.176	509.019	612.858	772.419	16	0	297.474	300.057	302.412	305.079	307.800	310.296	313.142
17	0	315.913	360.667	410.105	479.219	569.305	676.402	840.390	17	0	NA			NA			NA
Sum	237								Sum	168							

[a] LMS, lambda–mu–sigma.

Table 5
IGFBP-3 continuous reference values (ng/mL) using LMS[a] data obtained using the immunoradiometric assay (Immunodiagnostic Systems) system at Severance Hospital.

Boys	No.	Percentiles							Girls	No.	Percentiles						
		2.5	10.0	25.0	50.0	75.0	90.0	97.5			2.5	10.0	25.0	50.0	75.0	90.0	97.5
Age, years									Age, years								
0	2	782.476	1025.648	1175.411	1264.875	1360.079	1550.126	1980.750	0	7	806.069	1030.709	1176.153	1266.853	1365.382	1564.573	2038.708
1	26	842.168	1068.180	1232.709	1358.467	1495.530	1717.058	2139.936	1	31	880.073	1097.896	1259.084	1382.742	1520.020	1751.929	2224.205
2	43	916.538	1122.461	1294.369	1456.268	1636.822	1880.471	2280.100	2	40	966.360	1175.224	1347.520	1504.744	1682.571	1939.883	2396.281
3	141	1000.134	1186.830	1361.962	1558.389	1782.133	2041.549	2413.500	3	64	1065.503	1266.554	1447.988	1638.836	1857.966	2136.311	2574.396
4	292	1090.993	1262.324	1439.705	1668.170	1933.306	2206.203	2555.280	4	148	1172.897	1368.252	1558.119	1781.645	2041.251	2337.972	2760.347
5	390	1186.700	1352.768	1535.763	1791.128	2091.504	2381.253	2727.456	5	244	1279.055	1469.413	1666.550	1920.553	2218.133	2530.188	2937.825
6	435	1282.214	1455.215	1648.985	1924.906	2251.786	2564.179	2933.469	6	304	1380.973	1569.536	1774.217	2054.792	2385.317	2711.657	3111.792
7	457	1365.287	1554.212	1761.801	2050.603	2393.228	2730.481	3142.400	7	364	1465.193	1657.279	1870.022	2169.052	2521.893	2861.480	3266.399
8	481	1426.880	1635.610	1856.936	2150.979	2498.897	2858.184	3320.313	8	465	1514.556	1713.901	1933.272	2239.231	2599.640	2948.465	3367.052
9	532	1476.299	1697.880	1928.961	2229.145	2583.686	2958.099	3451.132	9	348	1536.800	1745.233	1969.631	2274.169	2631.609	2986.019	3422.658
10	426	1530.039	1752.592	1989.020	2303.937	2676.536	3060.121	3551.899	10	303	1582.248	1801.416	2033.481	2341.802	2702.600	3066.981	3524.770
11	455	1586.953	1811.199	2053.905	2385.180	2777.058	3169.122	3656.581	11	232	1671.084	1897.796	2139.857	2464.994	2845.559	3225.296	3696.367
12	280	1635.450	1873.774	2127.950	2468.358	2868.258	3271.589	3777.273	12	102	1770.744	2000.206	2250.989	2597.922	3004.832	3398.980	3872.594
13	150	1675.528	1943.866	2215.875	2556.340	2950.902	3369.301	3921.034	13	24	1853.948	2084.089	2341.138	2706.205	3135.103	3539.602	4011.528
14	62	1714.098	2026.230	2320.845	2654.607	3034.692	3470.137	4087.398	14	3	1920.217	2150.394	2411.906	2790.735	3236.253	3647.866	4117.022
15	21	1757.424	2123.235	2442.582	2765.774	3127.142	3579.247	4271.060	15	3	1977.908	2207.384	2472.251	2862.764	3322.382	3739.218	4204.204
16	5	1806.442	2234.069	2578.105	2887.163	3226.666	3693.110	4463.766	16	0	2032.384	2260.672	2528.355	2929.758	3402.520	3823.671	4283.558
17	2	1860.876	2358.522	2725.262	3015.406	3328.814	3805.157	4657.083	17	1	2085.465	2312.254	2582.451	2994.336	3479.748	3904.692	4358.921
Sum	4200								Sum	2695							

[a] LMS, lambda–mu–sigma.

Table 6
IGFBP-3 continuous reference values (ng/mL) using LMS[a] data obtained using the Immulite 2000 (Siemens) immunoassay system at Gangnam Severance Hospital.

Boys	No.	Percentiles							Girls	No.	Percentiles						
		2.5	10.0	25.0	50.0	75.0	90.0	97.5			2.5	10.0	25.0	50.0	75.0	90.0	97.5
Age, years									Age, years								
0	0	–			–			–	0	0	–			–			–
1	1	1510.632	1972.673	2411.959	2928.137	3474.773	3994.523	4609.401	1	0	–			–			–
2	1	1706.435	2194.019	2651.106	3182.263	3739.698	4266.364	4886.735	2	0	–			–			–
3	4	1919.689	2429.860	2901.399	3443.449	4007.592	4537.792	5160.564	3	6	2247.869	2795.400	3255.894	3742.773	4208.870	4613.725	5052.275
4	22	2165.177	2693.413	3174.703	3722.099	4287.381	4816.309	5436.581	4	13	2510.021	3038.104	3496.973	3994.505	4481.165	4911.195	5383.953
5	92	2453.324	2989.274	3470.411	4011.816	4566.816	5084.288	5690.944	5	65	2780.045	3292.613	3750.552	4258.671	4766.169	5222.468	5731.943
6	175	2761.739	3295.211	3767.034	4292.486	4827.627	5325.450	5909.943	6	152	3057.562	3555.597	4011.807	4529.476	5057.725	5541.660	6091.481
7	249	3063.371	3590.215	4049.262	4555.433	5068.117	5544.726	6106.397	7	229	3315.415	3798.589	4250.385	4773.412	5318.214	5826.945	6415.926
8	325	3346.279	3864.344	4308.754	4793.983	5283.210	5738.471	6278.336	8	278	3525.708	3996.901	4442.715	4965.983	5520.189	6047.022	6669.307
9	327	3544.128	4073.462	4520.021	5002.752	5487.691	5940.294	6481.982	9	277	3718.083	4196.326	4645.603	5173.522	5738.525	6286.037	6951.860
10	350	3625.543	4220.430	4712.803	5239.245	5766.411	6260.789	6859.882	10	208	3890.253	4420.710	4896.295	5441.779	6025.997	6607.708	7355.402
11	310	3689.026	4384.607	4947.036	5540.482	6132.673	6691.591	7379.688	11	135	4042.970	4667.505	5188.356	5761.041	6368.758	6990.064	7840.490
12	188	3835.599	4618.539	5233.975	5873.194	6508.746	7113.755	7873.757	12	58	4322.950	4944.288	5472.001	6053.447	6659.264	7255.001	8021.045
13	109	4014.482	4855.221	5494.329	6146.072	6791.929	7413.998	8215.945	13	24	4587.049	5171.667	5689.701	6267.700	6857.308	7406.365	8051.299
14	60	4196.807	5043.306	5661.858	6279.442	6889.907	7487.390	8283.828	14	9	4751.799	5341.375	5863.772	6441.200	7018.299	7539.828	8127.757
15	20	4365.320	5181.816	5750.987	6305.356	6852.473	7399.585	8160.832	15	0	4865.796	5473.868	6003.863	6579.760	7144.289	7644.299	8195.589
16	5	4495.079	5291.780	5816.435	6312.300	6801.196	7303.939	8042.441	16	1	4953.694	5583.933	6121.190	6693.839	7245.001	7725.364	8246.946
17	0	4595.265	5396.487	5888.770	6336.927	6778.567	7249.561	7990.834	17	0	5037.837	5690.851	6234.412	6802.714	7340.388	7802.418	8297.839
Sum	2238								Sum	2695							

[a] LMS, lambda–mu–sigma.

Table 7
IGFBP-3 continuous reference values (ng/mL) using LMS[a] data obtained using the cobas e801 (Roche) immunoassay system at Yongin Severance Hospital.

Boys	No.	Percentiles							Girls	No.	Percentiles						
		2.5	10.0	25.0	50.0	75.0	90.0	97.5			2.5	10.0	25.0	50.0	75.0	90.0	97.5
Age, years									Age, years								
0	0								0	0	–			–			–
1	2								1	2	1513.135	2016.321	2466.586	2966.862	3467.139	3917.403	4420.589
2	0	1993.104	2763.254	2763.254	2763.254	2763.254	2763.254	3910.795	2	2	1718.617	2214.621	2658.460	3151.596	3644.732	4088.571	4584.575
3	4	2218.250	2330.722	2573.814	3039.775	3594.841	3977.974	4185.480	3	2	1922.441	2411.365	2848.867	3334.964	3821.061	4258.563	4747.487
4	12	2375.469	2503.265	2757.708	3237.170	3795.909	4174.722	4393.909	4	6	2123.523	2605.468	3036.724	3515.882	3995.040	4426.296	4908.241
5	11	2440.749	2769.435	3092.850	3486.695	3922.163	4355.219	4889.200	5	5	2321.590	2796.655	3221.755	3694.073	4166.390	4591.491	5066.555
6	21	2552.053	3072.666	3439.723	3720.380	4017.495	4463.113	5240.380	6	13	2516.410	2984.693	3403.725	3869.300	4334.875	4753.907	5222.190
7	32	2752.367	3199.014	3564.975	3917.128	4290.156	4727.878	5347.855	7	20	2704.407	3166.004	3579.053	4037.981	4496.909	4909.959	5371.556
8	32	2910.910	3329.722	3691.038	4067.161	4461.926	4879.979	5420.972	8	32	2877.930	3332.936	3740.088	4192.463	4644.838	5051.990	5506.996
9	38	3063.546	3443.446	3794.126	4194.075	4609.833	5002.518	5461.655	9	34	3035.465	3483.974	3885.312	4331.227	4777.142	5178.480	5626.989
10	28	3259.851	3573.521	3905.009	4344.118	4796.341	5156.136	5512.544	10	26	3181.661	3623.765	4019.371	4458.919	4898.466	5294.072	5736.176
11	20	3434.414	3753.079	4087.554	4527.077	4974.193	5324.999	5668.234	11	14	3315.280	3751.070	4141.026	4574.296	5007.566	5397.523	5833.313
12	14	3561.909	3949.699	4303.985	4706.541	5111.275	5471.139	5869.104	12	6	3430.974	3860.539	4244.926	4672.008	5099.089	5483.476	5913.042
13	7	3687.233	4103.464	4461.995	4843.549	5223.633	5578.106	5986.347	13	4	3532.941	3956.370	4335.267	4756.248	5177.229	5556.125	5979.554
14	6	3863.437	4156.921	4477.110	4909.802	5336.434	5644.467	5921.158	14	1	3628.035	4045.416	4418.900	4833.868	5248.836	5622.320	6039.702
15	2	3996.433	4165.515	4451.215	4913.729	5364.640	5631.947	5786.231	15	0	3720.626	4132.039	4500.183	4909.218	5318.253	5686.397	6097.810
16	2	4023.188	4164.034	4439.688	4885.128	5315.442	5567.268	5691.867	16	0	3812.936	4218.394	4581.208	4984.321	5387.434	5750.249	6155.706
17	0	4025.835	4160.768	4424.347	4847.204	5252.465	5488.293	5604.333	17	0	3905.246	4304.748	4662.233	5059.424	5456.616	5814.101	6213.603
Sum	231								Sum	167							

[a] LMS, lambda–mu–sigma.